Bactericidal agents – Rifampicin (R), Isoniazid (H), Pyrazinamide (z), Streptomycin.

Bacteriostatic agents – Ethambutol (E). The regimen for treating TB consists of an intensive phase of 2 months of isoniazid, rifampin, pyrazinamide and ethambutol, followed by a continuation phase of 4 months of isonizid and Rifampin.

7.11.7 Prophylaxis and Control Measures

The BCG (Bacille – Calmette – Guerin) **administered** by intradermal injection of the live attenuated vaccine. The immunity may last for about 10 years.

The prevention of TB can be done by the following general measures such as

1. Adequate nutrition.
2. Practicing good hygiene (washing hands)
3. Health education.
4. Cover the mouth with a tissue when you cough or sneeze.

Infobits

Drug Resistance Tuberculosis

MDR-TB: Multidrug resistance tuberculosis refers to resistance to rifampian and Isoniazid. MDR-TB is a global problem especially in HIV-patients.

XDR-TB: Exensively drug resistance tuberculosis refer M.tuberculosis strains which are resistant to any fluoroquinolone and at lecest one of 3 injectable second line drugs (Kanamycin, Capreomycin and Amikarin), in addition to Isoniazid and rifampicin.



24 March is celebrated as World Tuberculosis day **Figure world TB day**

7.12 *Treponema Pallidum*

Treponema pallidum is included in the family *Spirochaetaceae*. They are slender spirochaetes with fine spirals having pointed ends. Some of them are pathogenic for humans, while others are normal flora in mouth and genitalia. These pathogens are strict parasites and the diseases caused by Treponema are called Treponematoses. *Treponema pallidum* is the causative agent of syphilis. The name *Treponema pallidum* is derived from **Greek** words, which means, **Trepos** – Turn **Nema** –Thread and **Pallidum** – Pale staining.

7.12.1 Morphology

It is thin, delicate spirochete with tapering ends, about 10µm long and 0.1–0.2 µm wide. It has about ten regular spirals, which are sharp and angular, at regular intervals of about 1µm. They are actively motile (endoflagella), exhibiting rotation around the long axis, backward and forward movements and flexion of the whole body. It cannot be seen under light microscope and does not take ordinary bacterial stains. It can be seen under the dark ground (Figure 7.22) or phase contrast microscope. It can be stained by silver impregnation method.

7.12.2 Culture

- Pathogenic Treponema cannot be grown in artificial culture media.

TN_GOVT_XII_Micro_Biology_CH07.indd 143 15-04-2019 16:38:38